

APTITUDE TEST (SET-B)

Instructions:

- All questions are compulsory
- Each question carries 1 mark

Section 1: Logical Reasoning (10 Marks)

1. Find the next number in the series:

7, 14, 28, 56, 112, ____

- A) 196
- B) 224
- C) 256
- D) 210

2. Find the missing term:

1, 3, 9, 27, 81, ____

- A) 162
- B) 243
- C) 324
- D) 200

3. In a certain coding system, each letter is replaced by its opposite letter ($A \leftrightarrow Z$, $B \leftrightarrow Y \dots$). What is the code for "BAT"?

- A) YZG
- B) YZG
- C) YZG
- D) YZG

4. Find the odd one out:

8, 27, 64, 125, 216, 300

- A) 27
- B) 125
- C) 216
- D) 300

5. Statements:

All cars are vehicles.

Some vehicles are electric.

Conclusions:

I. Some cars are electric

II. Some vehicles are cars

- A) Only I follows
- B) Only II follows
- C) Both follow
- D) None follow

6. Find the next term:

ZY, XW, VU, TS, ____

- A) RQ
- B) PQ
- C) RS
- D) OP

7. If in a code language, SUN = 1921 and MOON = 13151514, then how is RUN written?

- A) 182114
- B) 181421
- C) 182114
- D) 182114

8. A is the son of B. B is the sister of C. C is the father of D. How is A related to D?

- A) Uncle
- B) Brother
- C) Cousin
- D) Nephew

9. What is the angle between the hour and minute hand at 4:20?

- A) 10°
- B) 20°
- C) 30°
- D) 40°

10. If today is Thursday, what day will it be after 150 days?

- A) Sunday
- B) Monday
- C) Tuesday
- D) Wednesday

Section 2: Basic Mathematics (10Marks)

11. If $x + \frac{1}{x} = 6$, find $x^2 + \frac{1}{x^2}$

- A) 32
- B) 34
- C) 36
- D) 30

12. Solve the quadratic equation:

$$x^2 - 9x + 20 = 0$$

- A) $x = 4, 5$
- B) $x = 2, 10$
- C) $x = 1, 20$
- D) $x = 5, 6$

13. A train crosses a pole in 5 seconds and a platform of 150 m in 20 seconds. Find the speed of the train.

- A) 8 m/s
- B) 10 m/s
- C) 12 m/s
- D) 15 m/s

14. A sum becomes ₹1728 in 3 years at 20% compound interest. Find the principal.

- A) 800
- B) 900
- C) 1000
- D) 1200

15. A and B together can do a work in 10 days. B alone can do it in 15 days. In how many days can A alone do it?

- A) 20
- B) 25
- C) 30
- D) 35

16. Find the remainder when 2^{50} is divided by 3.

- A) 0
- B) 1
- C) 2
- D) 3

17. If $\cos \theta = 12/13$, find $\sin \theta$.

- A) $5/13$
- B) $12/5$
- C) $13/12$
- D) $5/12$

18. A person travels equal distances at speeds of 40 km/h and 60 km/h. Find the average speed.

- A) 48 km/h
- B) 50 km/h
- C) 52 km/h
- D) 45 km/h

19. How many distinct arrangements can be made using the letters of the word "MISS"?

- A) 12
- B) 24
- C) 6
- D) 8

20. The ratio of two numbers is 5:7 and their sum is 96. Find the smaller number.

- A) 35
- B) 40
- C) 45
- D) 50

Section 3: DSA (10 Marks)

21. What is the average-case time complexity of linear search?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

22. Which data structure is used in Depth First Search (DFS)?

- A) Queue
- B) Stack
- C) Array
- D) Tree

23. What is the time complexity of searching in a hash table (average case)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

24. What will be the output of the following code?

Java

```
int a = 2;
```

```
int b = 3;
```

```
System.out.println(a++ * b++ + ++a);
```

- A) 8
- B) 9
- C) 10
- D) 11

25. Which traversal technique uses recursion by default?

- A) BFS
- B) DFS
- C) Level Order
- D) None

26. What is the worst-case time complexity of insertion sort?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n^2)$
- D) $O(n \log n)$

27. What will be the output?

Java

```
System.out.println("10" + 5 + 5);
```

A) 1055

B) 20

C) 155

D) 105

28. What is the best-case time complexity of linear search?

A) $O(1)$

B) $O(n)$

C) $O(\log n)$

D) $O(n^2)$

29. What is an AVL tree?

A) Unbalanced tree

B) Self-balancing BST

C) Heap

D) Graph

30. Which data structure is used to implement priority queue?

A) Stack

B) Queue

C) Heap

D) Linked List